# 4 Multiple Integrals

## 4.1 Double Integrals over Rectangles

**Definition 4.1.1** The double integral of the function $f$ over the rectangle $R$ is defined to be

$$\iint_R f(x,y)\,dA = \lim_{m,n\to\infty} \sum_{i=1}^{n}\sum_{j=1}^{m} f(x_i,y_j)\Delta A,$$

if this limit exists.

**Example 4.1.2** Approximate the volume of the region above the $x,y$-plane and below the surface $z = 16 - x^2 - 2y^2$ over the region $R = [0,2]\times[0,2]$.

**Theorem 4.1.3** *(Fubini's Theorem) If $f$ is continuous on the rectangle $R = [a, b] \times [c, d]$, then*

$$\iint_R f(x, y)\, dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy.$$

**Example 4.1.4** Evaluate the iterated integral

$$\int_0^2 \int_1^2 (x - 3y^2) dy dx.$$

**Example 4.1.5** Evaluate

$$\iint_R y \sin(xy)\, dA$$

where $R = [1, 2] \times [0, \pi]$.

**Example 4.1.6** Find the exact volume in the first example. That is, find the volume beneath the surface $z = 16 - x^2 - 2y^2$ over the rectangle $R = [0, 2] \times [0, 2]$.

### 4.1.1 Average Value

**Definition 4.1.7** The average value of the function $f(x, y)$ over the rectangle $R$ is defined by

$$f_{avg} = \frac{1}{A(R)} \iint_R f(x, y)\, dA,$$

where $A(R)$ is the area of the rectangle $R$.

**Example 4.1.8** Find the average value of the function

$$f(x, y) = x^2 - 2y^2$$

over the rectangle $R = [-1, 1] \times [-1, 1]$.